

| Seat system size                              | Size 1<br>Standard | Size 1<br>Longer | Size 2<br>Standard | Size 2<br>Longer | Size 3<br>Standard | Size 3<br>Longer |
|-----------------------------------------------|--------------------|------------------|--------------------|------------------|--------------------|------------------|
| Seat depth min / max                          | 19cm /<br>30cm     | 26cm /<br>37cm   | 26 / 37cm          | 34cm /<br>37cm   | 34cm /<br>45cm     | 41cm /<br>52cm   |
| Pelvic width min / max                        | 21cm /<br>32cm     | 21cm /<br>32cm   | 25cm /<br>36cm     | 25cm /<br>36cm   | 30cm /<br>41cm     | 30cm /<br>41cm   |
| Thoracic width min /<br>max                   | 19cm /<br>36cm     | 19cm /<br>36cm   | 25cm /<br>40cm     | 25cm /<br>40cm   | 31cm /<br>46cm     | 31cm /<br>46cm   |
| Sacral pad height                             | 10cm               | 10cm             | 12.5cm             | 12.5cm           | 15cm               | 15cm             |
| Minimum backrest<br>height<br>(starting from) | 30cm               | 30cm             | 37.5cm             | 37.5cm           | 47.5cm             | 47.5cm           |
| Maximum User weight                           | 136kg              | 136kg            | 136kg              | 136kg            | 136kg              | 136kg            |

### Liability

RMS recommends using the device under the specified conditions and for the intended purpose. The device must be maintained according to the instructions for use supplied with the device. The manufacturer is not liable for any adverse outcome caused by any component combinations that were not authorized by them.

### CE Conformity

This product meets the requirements of MDD93/42/EEC for Medical Devices and MDR 2002 (2002 No. 618). This device has been classified as a Class 1 Medical Device according to the classification criteria outlined in the regulation. The Declaration of Conformity is created by RMS as a conformation of compliance.

### Reporting of Adverse Incidents

In the unlikely event of a serious adverse incident occurring, please report to the manufacturer and the National Competent authority (MHRA in UK)



## Virtus Seating System (VIRTUS-MSS)

### Instructions For Use

These Instructions are intended for use by a Healthcare Professional, Carer or Approved Repair agent. The document will highlight any section which can only be performed by a healthcare professional. Please read in full and ensure you understand all instructions for use, in particular all safety information and maintenance instructions. For VIRTUS Seat units 01/2020 to current.

VIRTUS-MSS-IFU-2022- rev0

**Application:** This Virtus seating system is to be only used in combination with approved wheelbases / wheelchairs and interface systems. Any third party accessories must only be used in conjunction with a risk assessment performed by the healthcare professional.

#### Medical Benefits:

The Virtus Seating System is designed to meet the needs of wheelchair users who have an inability to maintain a symmetrical and unsupported sitting posture. The Virtus has a dynamic backrest option which permits the user to extend safely, whilst ensuring their posture is returned to neutral.

The Virtus is fully adjustable allowing the seat to grow with the individual, thereby continuing to meet postural needs.

#### Contents:

- |                                                         |                                                 |
|---------------------------------------------------------|-------------------------------------------------|
| 2. Safety guidelines.                                   | 9. Pelvic support and Knee adductor             |
| 3. Safety guidelines and Tools required.                | 10. Seat cushion depth and Footrest adjustment. |
| 4. Interface fitment.                                   | 11. Footplate adjustment.                       |
| 5. Virtus seat unit fitment                             | 12. Armrest adjustment.                         |
| 6. Backrest angle adjustment                            | 13. Pommel, Headrest and Tray.                  |
| 7. Sacral pad and Dynamic backrest bearing replacement. | 14. Upholstery Care.                            |

#### Safety Guidelines and Pre-use Checks

Only a Healthcare Professional can prescribe and setup the Virtus for an occupant.

Prior to commencing the installation of a Virtus Seating System, the surrogate wheelbase should be placed on a level non-slip surface with the parking brakes applied.

For correct support and user comfort. Initial installation and adjustments must be carried out by a suitably qualified person.

It is the responsibility of the installer to ensure that all Interface Mounting and Component Screws and Nuts are correctly tightened, before handover to the user. (This does not apply to Virtus Seating Systems interfaced with a surrogate base at the RMS factory).

The Virtus Seating System has a Maximum User weight of 136kg.

As the installation of a Virtus Seating System may raise or position the user further forward in their wheelchair, an appropriate stability test should be carried out prior to final commissioning.

After the Virtus Seating System has been initially adjusted to suit the individual user, any settings should not be subsequently affected by the removal and refitting of the seat unit. However, care should be taken not to impact adjustable components whilst the seat unit is removed from the wheelchair as this could affect their original pre-set position.

To accommodate any changes in user growth or postural positioning, it is recommended that the user be checked at regular intervals by a Healthcare Professional, to ensure that adjustable components are correctly set to suit the user's current requirements.

## Transportation

The **Virtus** Seating System has been successfully Crash Tested to ISO 16840-4 (Seating Devices for use in Motor Vehicles) in the forward facing position with a 50<sup>th</sup> Percentile 75kg ATD, interfaced with an approved surrogate test base. However, this does not imply and is not intended to imply in any way, that the **Virtus** Seating System is suitable for transporting an occupant in a motor vehicle.

The **Virtus** Seating System must be correctly interfaced with a host wheelbase of a type and model which has been successfully Crash Tested and meets the requirements of ISO 7176-19.

The transporting vehicle MUST also be suitably equipped with wheelchair and occupant restraint systems compliant with ISO 10542-1: 2012.

Please follow the instructions for use in transportation provided by the wheelchair and restraint manufacturer.

**Under No Circumstances should any part of a transportation restraint system, be attached directly to the Virtus Seating System.**

Whilst provision is made on the **Virtus** Backrest for attaching a postural positioning belt or harness, which should still be utilised as normal during the transportation period, these support devices should **NEVER** be used as the sole method of occupant restraint during transportation.

#### Risk analysis for Wheelchair Transportation:

A comprehensive risk assessment is advisable, taking into account the user's requirements and the type of transport they plan to use. Potential risk areas (not limited to):

The user being transferred to a fixed vehicle seat.

The wheelchair/seating unit being transported as unoccupied luggage.

The user occupying a wheelchair/seating unit during transportation.

Ability of the WTORS to hold the wheelchair and/or seating unit together with the user during vehicle movement or in the event of an impact.

Effects of normal vehicle manoeuvres, such as braking, accelerating and cornering on the wheelchair user.

Effects of the occupant restraint on the user, both in normal vehicle movement and on impact.

Effects of the seating unit, wheelchair and / or any accessories being used, on the action of a vehicle anchored occupant restraint in an impact.

Suitability of any interface that connects the seating unit to the wheelchair.

Postural support or belt/harness that is not sufficiently strong to withstand the force of an impact.

Requirements for the use of a headrest to restrict the rearward movement of the head during vehicle motion or impact.

The effect on other passengers if the user, wheelchair/seating unit or accessories were to become detached during impact.

More Information on Transportation and Risk Assessment :

[www.pmguk.co.uk](http://www.pmguk.co.uk), [www.gov.uk](http://www.gov.uk),  
[www.hse.gov.uk/risk/assessment](http://www.hse.gov.uk/risk/assessment).



## Upholstery Care and Maintenance (All Users)

Unless otherwise requested, your **Virtus** Seating System will be finished in 100% Polyester, 3mm "Spacer" fabric.

Alternative surfaces may be textured or smooth (reversed).

Zip-removable covers enable easy removal, for cleaning or machine washing.

The "Spacer" fabrics have the following qualities:-

Flame Retardant to BS5852-2006 Ign. source 5 Crib 5 and meet or exceed BSEN1021-1&2 Anti-Bacterial / Anti Fungal i.e. will not support microbial or fungal growth including Salmonella, E Coli and MRSA. Waterproof — Water or oil, Breathable — Moisture vapour permeable.

Plastic components—use a water dampened cloth and dry afterwards.

### General Fabric Care

Use of a vacuum cleaner or soft brush, to remove dust and solid particles, together with frequent wiping with a damp cloth, will help to maintain the appearance of the fabrics.

**Spillages** should always be removed as quickly as possible with an absorbent dry cloth.

**Minor Soiling** can normally be removed with a clean damp cloth.

**Water Based Stains** can be cleaned using mild liquid detergent with warm water. Use cold water for stains produced by bodily fluids, then rinsed thoroughly, followed by drying with a clean absorbent cloth.

**Light Steam Cleaning** can be carried out in situ, ensure all areas are completely dry before reuse.

**Machine Washing** with care @ 40°. Covers must **NOT** be turned inside-out and any zips should be fully closed. Remove from machine as soon as the wash cycle has finished.

### DO NOT DRY CLEAN

**Line Dry** ideally the upholstery should be "Line Dried" do **NOT** Tumble Dry.

See Symbols below.

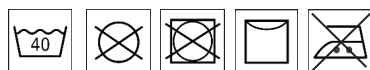
All soap residue must be thoroughly rinsed out of all fabrics, as it can attract further stains.

All stains caused by bodily fluids, should be removed immediately.

As with all fabrics, stains that are left in situ for over 24 hours will have set and will become increasingly difficult to remove.

Always pre-test cleaners on a hidden area of fabric to test for colour-fastness and texture before proceeding with cleaning.

Failure to adhere to the above instructions could void any warranty.



### Seat Unit Servicing :

The Virtus Seat unit should be serviced annually, however in heavy use this should be reduced to six monthly intervals or less.

**Disposal:** If the Virtus is no longer required, it must be safely disposed of.

Remove any labels. Remove any parts that can be dismantled to reduce the risk of re-use by unintended users. Ensure the Healthcare Professional is aware the device is no longer required. Be careful of any sharps and wear gloves. Do not reuse any components.

Do not dispose of the seat unit in household waste. Check your local recycling centre for disposal of Foams, Material, Aluminium and Polypropylene.



Carers should ensure correct utilisation of any positioning Straps or Harnesses, as failure to do so could result in injury to the user. It is recommended that any postural Straps or Harnesses being used, are the first items to be secured when the user enters the seat and the last items to be released before exiting.

Worn or damaged components can lead to hygiene contamination, cause injury to the user and, in some cases, fail to support the user correctly. Regular inspections of all upholstery should be made and any defects should be reported to the relevant authority for rectification as soon as possible.

Ancillary devices, such as the Headrest, positioning Straps and Harnesses, or Pommel should be checked for security on a daily basis.

The wheelchair parking brakes should always be applied before attempting to transfer the occupant, removing or refitting the Virtus Seating System or making any adjustments.

Never hang heavy objects on the Virtus Seating System or any part of the wheelchair, as this could seriously affect the overall stability.

Always keep the Virtus away from sources of heat such as naked flames, cigarettes, electric and gas heaters.

Due to its weight, care should be taken when fitting and removing the Virtus seat unit and the host wheelchairs brakes should always be applied. To help reduce weight before lifting all accessories should be removed.

Do not place heavy objects on the tray and never move the seat / wheelchair with hot liquids / substances on it.

When fitting the Virtus onto its interface please ensure it is locked onto the interface correctly by pushing and pulling the top of the backrest. The seat should not lift at the front or slide forward.

Should the backrest resistance become less effective and need increasing / decreasing the dynamic rubber may need changing page 7.

### IFU and Label Symbols

|  |                     |     |                |  |                            |  |                          |  |                          |
|--|---------------------|-----|----------------|--|----------------------------|--|--------------------------|--|--------------------------|
|  | Date of manufacture | SN  | Serial No.     |  | Caution                    |  | Use Tools                |  | UK Conformity Assessment |
|  | Manufacturer        | REF | Catalogue No.  |  | CE Mark EU Conformity      |  | Single User Multiple Use |  |                          |
|  | Lot / Batch no.     | MD  | Medical Device |  | Consult Instruction Manual |  | EC Representative        |  |                          |

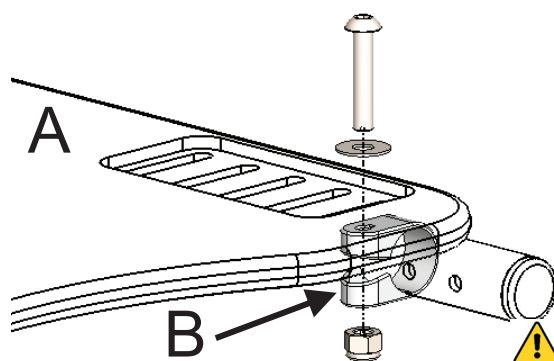
**Only a trained, approved Healthcare Professional should perform the fitting and/or setting up of any medical device. If you are uncertain about the function of the device please contact your healthcare professional for advice. Everyday use sections can be performed by all users.**



### Tools required for installation and adjustment:

|                       |                              |
|-----------------------|------------------------------|
| 3mm Hex key           | 10mm spanner                 |
| 4mm Hex key           | 13mm spanner                 |
| 5mm Hex key           | 8mm drill bit                |
| Drill countersink bit | Medium Flat head screwdriver |

## Adjustable Virtus Interface (Healthcare Professional Only)



The Virtus interface **A** can be fitted to a host wheelchair such as an Ot-tobock Discovery using the round tube clamps and nuts and bolts **B**.

The clamps can be positioned around the seat frame and secured using the bolts supplied picking up an appropriate slot in the interface. The frame clamps may need to be angled slightly to keep the board flat.

Torque setting: 15 Nm.

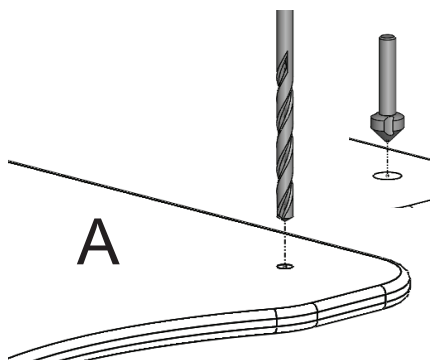
Ensure all four clamps are fully secure and the interface board is captive before use.



## Blank Virtus Interface (Healthcare Professional Only)

Where this type of attachment is not suitable it may be necessary to drill and bolt the the interface board **A** (blank interface boards available). We advise that four M8 high tensile countersunk bolts be used, two front and two back (a minimum of 20mm from any edge and any hole centres should be 20cm apart.) and this is secured to suitable structurally secured point on the wheelchair using nyloc nuts and large M8 washers. In order to allow the Virtus seat to sit correctly the M8 hole should be countersunk on the top of the interface.

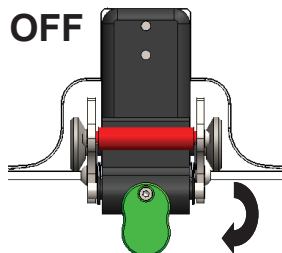
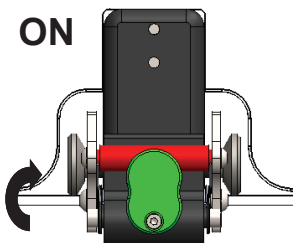
Torque setting:



Ensure all four bolts are fully secure and the interface board is captive before use. Over countersinking can weaken the structure of the board.

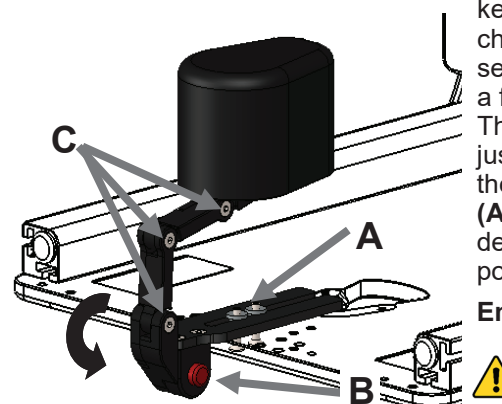
## Positive Locking Interface Mount (All Users)

When the seat stem is located into the positive locking interface mount the green safety locking lever can be rotated into the on or off position



**!** The Green safety locking lever should always be in the ON (locked) position before use, failure to so could result in an incident to occur.

## Flip Down Pommel (optional) (Healthcare Professionals only)



The pommel is fitted using two bolts and key washers **A**. Remove the seat cushion. The key washers sit in the pommels slotted channel and are bolted from underneath the seat. Leave a 25mm gap to the seatboard if a footrest is fitted.

The pommels height and angle can be adjusted to suit by loosening bolts **C** adjusting the pommels arms and re-tightening.

**(All Users)** To flip the pommel out of the way depress the Red or Green button **B**, the pommel will lock automatically when raised.

Ensure the pommel is secure before use.



## Headrest (optional) (Healthcare Professional Only)

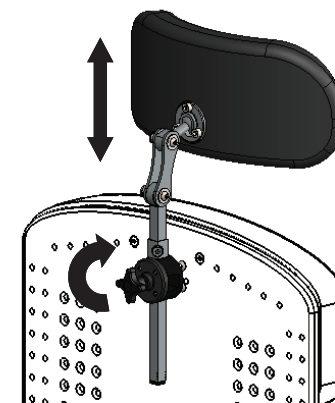
The headrest boss is mounted onto the pre-drilled holes on the Virtus backrest using bolts supplied. It has two positions depending on the size of the headrest stem, either 13mm or 16mm.

**(All Users)** The headrest can be inserted into the boss and adjusted to suit and held in place by turning the thumbscrew on the mounting boss.

Holes are pre-drilled for other RMS headrest fittings.

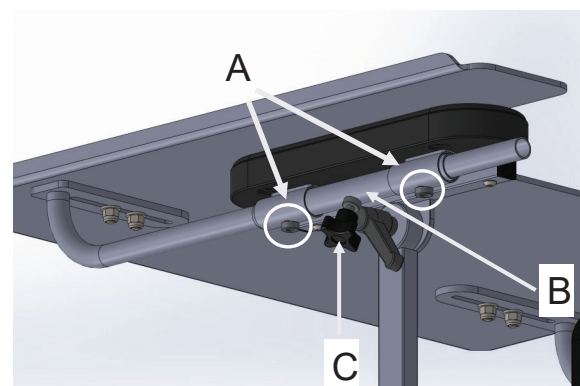


Ensure headrest adjusted correctly and tightened before use.



## Tray (optional) (Healthcare Professional Only)

To fit the tray mounts, undo the two arm pad screws and remove the arm pad from the channel. Slide tube **B** between the two receiver brackets **A**, line the slots in **A** with the arm pad screw holes between the armpad and the channel and reinsert the screws and tighten into the armpad.



Adjust the angle of tube **B** to allow easy access to thumb screw **C** when the tray is attached and tighten the grub screws circled.

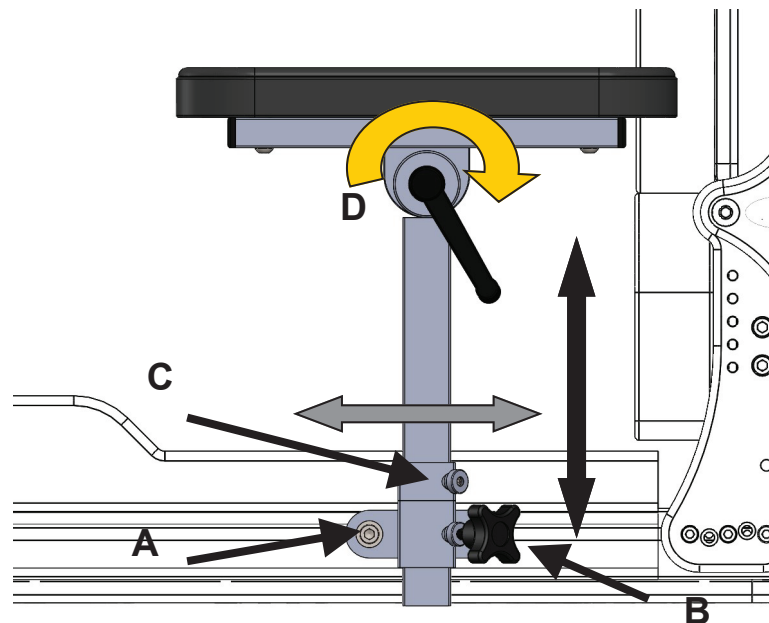
**(All Users)** Slide the tray brackets into receiver **B** and once in position tighten the thumb screw **C** on both sides.

Ensure tray and mounts are secure before use. Do not put heavy items on the tray.





## Armrest Adjustment (optional) (Healthcare Professional Only)



The armrest position can be adjusted by:

- 1) Loosening bolts **A** and sliding the mounting bracket to the desired position and retighten bolts **A**. 10Nm

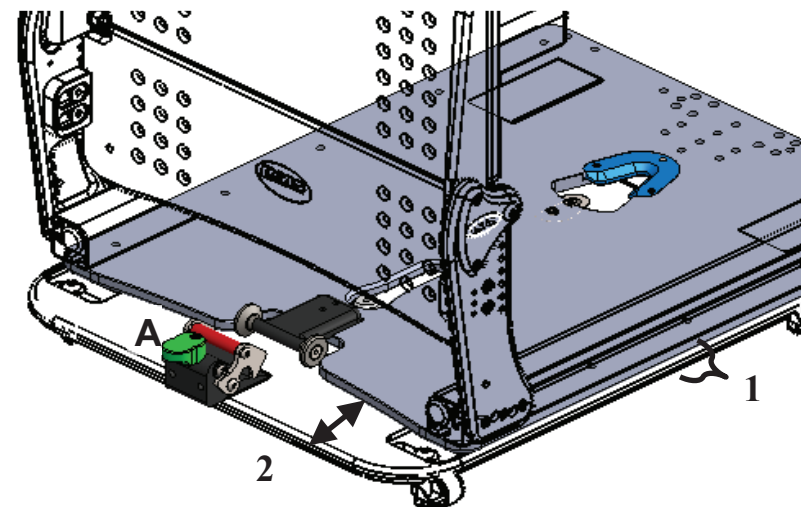
Armrest height adjustment:

- 1) Loosen hand knob **B** and the grub screw in memory collar **C**, adjust the armrest height to the desired position, retighten the hand knob **B** and slide the memory collar **C** to the armrest mount and retighten the grub screw.

To adjust the armpad angle, loosen lever arm **D** and adjust the armpad angle and retighten the lever arm **D**.

**⚠ Ensure Arm rest is correctly adjusted for height, position, angle and tightened correctly before use.**

## Virtus Seat Fitment (All Users)



The positive locking interface mount lever **A** should be set into the unlocked position (see page 4).

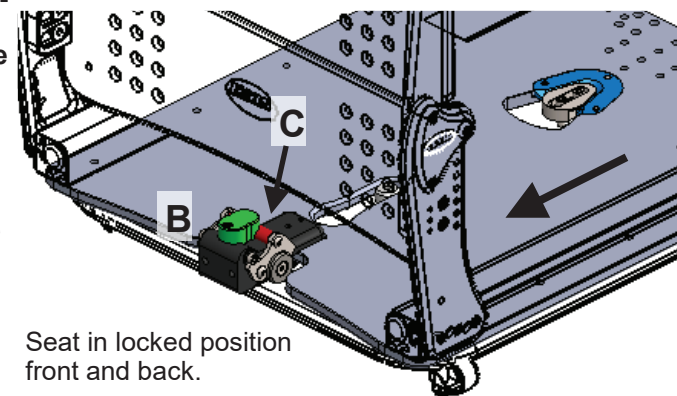
Place the Virtus onto the interface making sure sides of the seat are parallel with the side of the interface **1** and leave approximately a 3" gap between the back of the board and the back of the interface **2**.

Slide the seat unit forwards until the interface locking mount catch engages with the seat locking stem and rotate the green interface locking lever into the locked position **B**.

To Remove the Virtus seat, unlock the green safety lever **A** and pull the red bar **C** upwards on the positive locking mount and push the Virtus forward until it is clear of the mount. The Virtus can now be lifted off the interface.

**⚠ Care should be taken when lifting the Virtus seat unit due to its weight, removal of accessories such as the footplate will lighten the unit. Please ensure the wheelchairs brakes are on.**

**⚠ To check that the front and rear of the seat is correctly located, push and pull the top of the backrest. The seat should not lift at the front or slide forward.**



Seat in locked position front and back.

## Adjusting Backrest Angle and Seat Depth (Healthcare Professional Only)

### Backrest Angle Adjustment

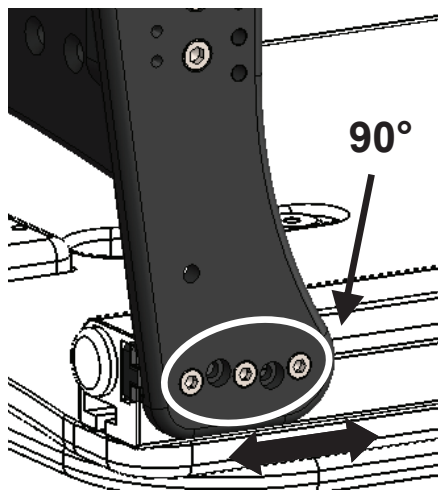
The back rest angle can be set at either 90° or 100°

To alter the backrest angle, loosen the centre of the three bolts and remove the outer bolts on both sides, adjust to the required angle and replace the bolts in the positions shown and retighten all three.

Torque setting: 20Nm

**⚠ Ensure the backrest is at the same angle both sides.**

**⚠ Check the backrest is secure before use.**



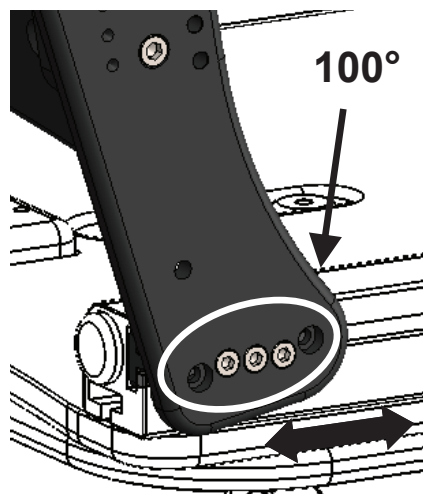
### Seat Depth Adjustment

The depth of the seat can be altered by loosening all three bolts on both sides and sliding the backrest assembly forwards or backwards to the desired depth and retightening the bolts.

Torque setting: 20Nm

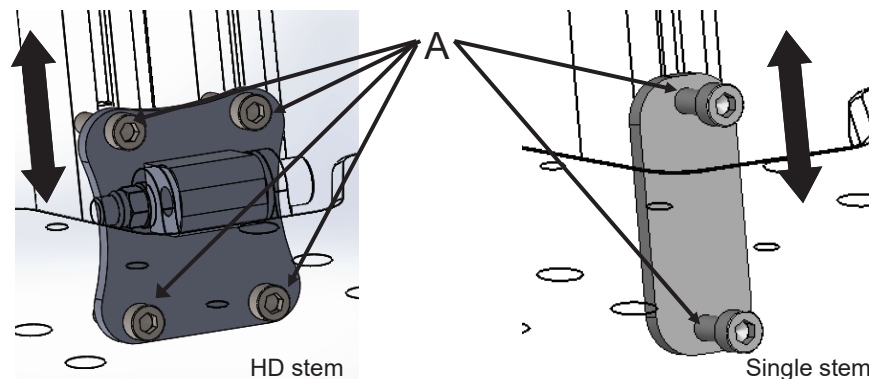
**⚠ The backrest assembly must not be extended beyond the seat side extrusion, the maximum rearwards travel is shown in the illustrations above and right.**

**⚠ Ensure the backrest is secure before use.**



\* The fixed Backrest Virtus has the same angle and depth adjustment as the Dynamic version above.

## Footplate Adjustment (Healthcare Professional Only)

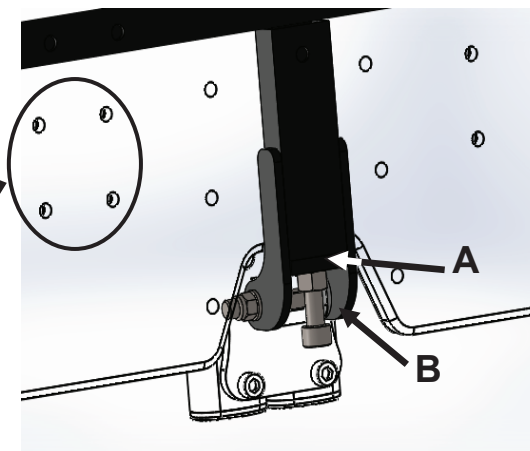


To adjust the height of the footplate loosen the upper and lower bolts **A** on the footplate mount. Slide the footplate up the stem and retighten. Torque setting: 20 Nm

**⚠ Ensure the footplate is adjusted correctly and fully tightened before use.**

The fixed and flip-up footplate board can have its depth adjusted by releasing the three bolts that hold the board onto the footboard bracket and moving the board to an appropriate pre-drilled hole (circled).

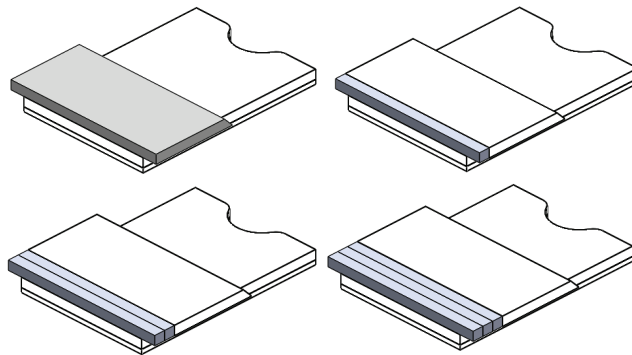
Individual footplates have pre-drilled holes to alter depth and are held onto the footboard arm by individual clamps, which can be released to make adjustment to depth and angle and retightened. Torque setting: 20 Nm



To adjust the angle of the flip up footplate, loosen nut **A** and turn bolt **B** in the desired direction until it sits at the correct angle and retighten nut **A**.

**⚠ Ensure all footplates fixings are secure before use.**

## Seat Cushion Ramp Depth Adjustment (Healthcare Professional Only)



The Virtus is supplied with 2 (size 1) or 3 (size 2/3) evazote foam infill sections to allow for seat ramp depth adjustment. To fit these turn the seat cushion over and unzip the seat cover. Remove the evazote cushion assembly, move the main ramp section adding or removing the infill sections to

suit (these are held in place with hook and loop). The seat assembly can then be reinserted into the cover a zipped up.

**! Please ensure the cushion assembly is adjusted correctly and reassembled into the cover correctly. After adjustment it is advisable check the cushion is suitable for the user .**

## Footrest (optional) (Healthcare Professional Only)

The Virtus footrests are available in two stem and three footplate options.

Fitment to the Virtus seat is the same for both the single and double stems.

Hold the stem at slight rearward angle at the top. Line up the four locating pins with the slots in the mount and push backwards, then exert a slight downwards force until the stem drops into place. (see Figs 1 & 2 )

The security pin can then be place through the monut and stem.

**! Ensure the footrest is locked in place correctly before use and security pin is correctly fitted.**

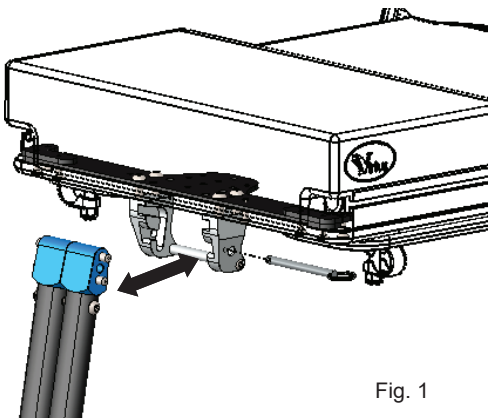


Fig. 1

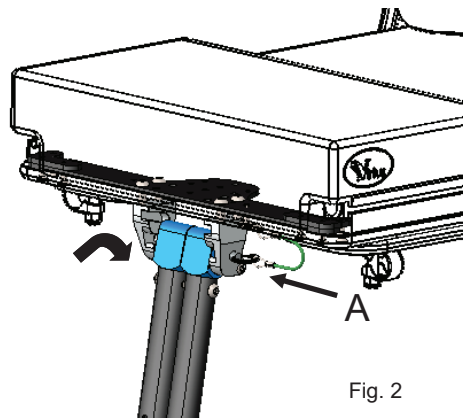
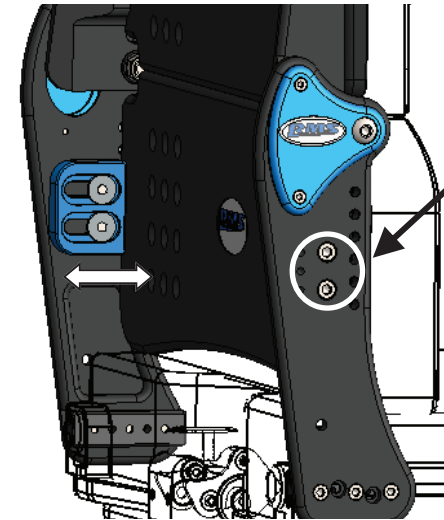


Fig. 2

## Sacral Panel Depth Adjustment (Healthcare Professional Only)



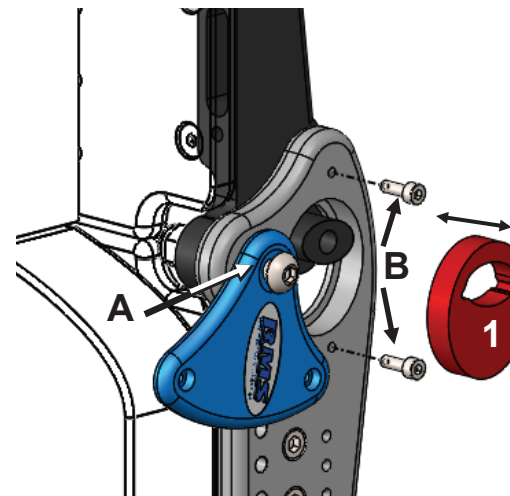
Loosen the bolts on both sides (arrowed) and slide the sacral panel along the brackets to the desired depth.

Check both sides are set at the same depth and retighten the bolts.

Torque setting: 18Nm

**! Ensure the sacral panel is secure before use.**

## Dynamic Rubber Replacement (Healthcare Professional Only)



To change the dynamic rubber 1, loosen bolt A and remove bolts B

Swing the Blue cover forward, pivoting on A.

Hold the backrest and carefully remove the bearing rubber.

**! Caution as the backrest could tilt slightly and trap fingers!**

Carefully insert the replacement bearing rubber by lining the backrest and the bearing guide pin. Swivel the blue cover plate back and reinsert bolts B and retighten A and B.

Torque setting: 15 Nm

**! Ensure the cover plate is tight and the backrest is sitting correctly on both sides.**

The Dynamic Rubbers are available in three resistance levels.

For identification, these are colour coded as follows:-

**Blue DX** Low Resistance, **Black DY** Medium Resistance, **Red DZ** High Resistance

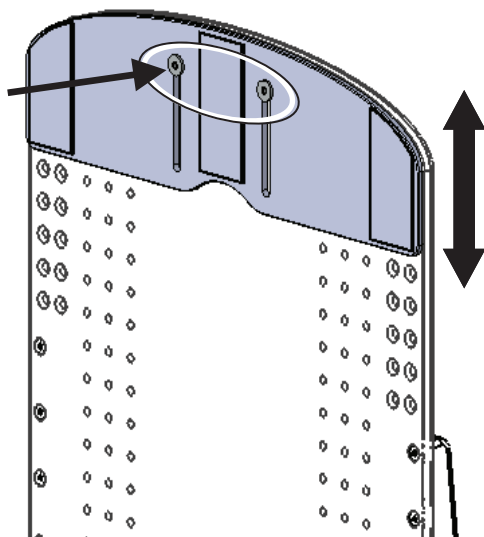
## Backrest Height Adjustment (Healthcare Professional Only)

To adjust the height of the backrest extender, loosen the two bolts that hold it in position slide the panel to the desired position and retighten both bolts.

Torque setting: 15Nm

The backrest cushion can be moved to suit and the gap left between cushioning taken up by inserting the extension infill cushion supplied with the Virtus.

**Ensure the extender panel is secure before use.**



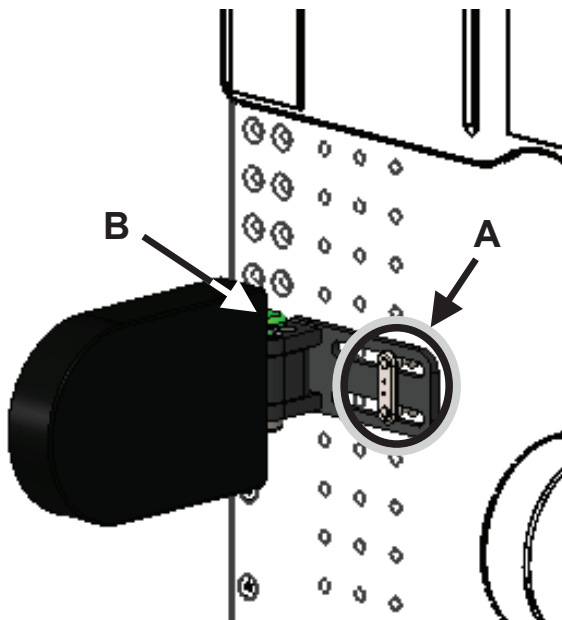
## Lateral Thoracic Support Adjustment (All Users)

To adjust the width of the thoracic loosen the two bolts **A** from the rear of the backrest and slide the thoracic to the desired width and re-tighten.

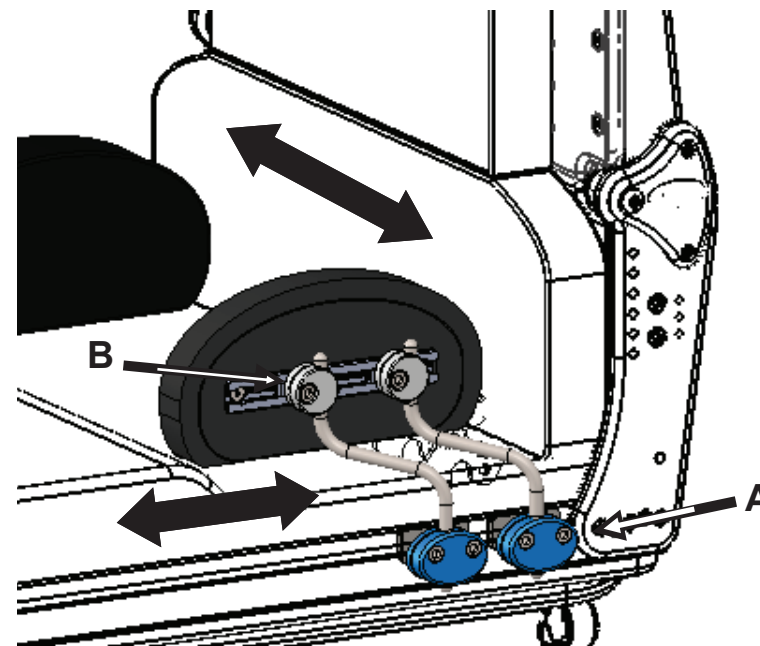
For greater adjustment the backrest has a number of pre drilled holes that allow the thoracic to be raised and lowered accordingly.

Swing away thoracic versions can be swung to one side by depressing the green button **B**. when swung back into position they automatically lock back .

**Ensure the thoracics are adjusted correctly and secure before use.**



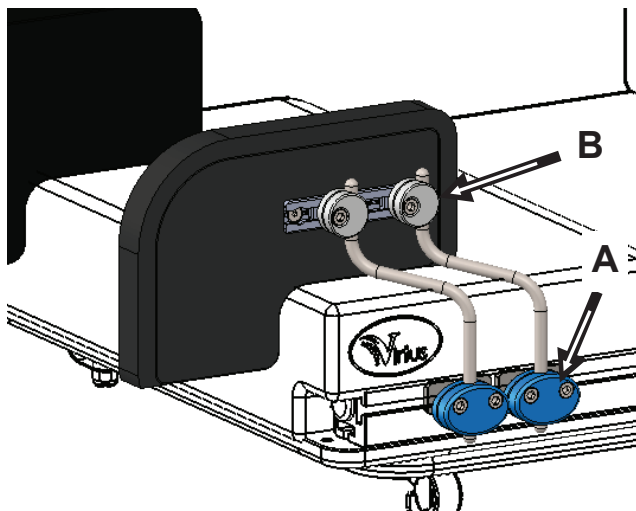
## Pelvic Support Adjustment (Healthcare Professional Only)



Loosen the bolts on clamps **A** and **B** and adjust the width and the depth of the pelvic supports to suit, allowing the arms to rotate in clamps **A** & **B**. Retighten the clampsonce set into the desired position.

Torque setting: 18Nm

**Depth adjustment is available on both clamps A or B.**  
**Ensure correct adjustment and security of pelvic supports before use**



## Knee Adductor (Healthcare Professional Only)

The Knee Adductors can be adjusted in the same way as the Pelvic supports (above) by loosening **A** & **B**, adjusting to suit and re-tightening.  
Torque setting: 18Nm



**Ensure correct adjustment and security of knee adductors before use.**